

POWER BI BUSINESS ANALYTICS PROJECT

SUPERSTORE BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

Interactive Sales • Profitability • Customer • Product Dashboard

KEY METRICS	VALUES
Total Sales	\$2M
Total Profit	\$286K
Total Orders	5K
Total Customers	793
Profit Margin	12.5%
Average Order Value	\$459
Profit YoY	14.2%

Personal Data Analytics Portfolio Project
Microsoft Power BI • Power Query • DAX

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2. Executive Summary

This project presents an interactive Business Intelligence solution built in Microsoft Power BI from the Sample Superstore transactional dataset. The purpose is to convert transactional data into a management-oriented view of sales, profitability, customers, products, regions, discounts and shipping modes.

The completed dashboard contains four analytical pages—Sales, Profitability, Customers and Details—with KPI cards, slicers, trend analysis, category and regional comparisons, profitability analysis, customer/product rankings and detailed tables. The documented dashboard reports approximately \$2M in sales, \$286K in profit, 5K orders, 793 customers, a 12.5% profit margin and a \$459 average order value.

The strongest portfolio-level finding is that revenue volume and financial quality are not identical. Technology leads both sales and profit, while Furniture generates substantial sales but considerably less profit. The West region is the strongest displayed region, Consumer is the largest customer segment, and Standard Class is the dominant shipping mode by sales.

3. Project Objectives

Objective	Purpose
Measure overall performance	Track sales, profit, orders, customers, profit margin and average order value.
Analyze sales drivers	Compare performance across categories, regions, customer segments and shipping modes.
Analyze profitability	Identify strong and weak sub-categories and products, including negative-profit items.
Understand customers	Compare Consumer, Corporate and Home Office segments and rank individual customers.
Support product decisions	Evaluate product sales, profit, quantity and discount together.
Enable self-service analysis	Use slicers, navigation, filters and cross-filtering to support interactive exploration.

4. Business Problem / Problem Statement

Retail transaction data can contain large amounts of information but still make it difficult to see what is driving business performance. The project addresses this reporting gap by organizing the available sales, profit, customer, product, geographic, discount and shipping information into an interactive dashboard.

Key business questions

- How are sales and profit performing overall?
- Which categories and sub-categories contribute most to sales and profit?
- Which regions generate the strongest sales?
- Which customer segments and individual customers contribute most value?
- Which products generate strong sales and which products create losses?
- How does discounting appear alongside profitability?
- How do sales change over time?
- Which shipping mode dominates sales?

5. Dataset Overview

Dataset attribute	Project detail
Dataset	Sample Superstore
Source/type	CSV / transactional retail dataset as documented in the project README
Records	9,994
Fields	21
Period	2014–2017
Business domain	Retail / Superstore
Granularity	Transaction / order-line level
Dimensions	Time, customer, segment, geography, product, category, sub-category, ship mode
Numerical fields	Sales, Quantity, Discount, Profit

The dataset fields documented in the project include order and shipping information, customer information, geographic fields, product hierarchy fields and four principal business variables. This structure supports descriptive analysis across time, geography, customers, products and financial performance.

6. Data Dictionary

Field name	Data type	Description	Business purpose
Row ID	Whole number	Row-level identifier	Record identification
Order ID	Text	Order identifier	Order/order-line analysis
Order Date	Date	Date the order was placed	Time-series analysis
Ship Date	Date	Date the order was shipped	Shipping/time analysis
Ship Mode	Text	Shipping method	Shipping mix analysis
Customer ID	Text	Customer identifier	Customer counting and analysis
Customer Name	Text	Customer name	Customer ranking/detail
Segment	Text	Customer segment	Segment comparison
Country	Text	Country	Geographic context

City	Text	City	Geographic detail
State	Text	State	Geographic detail
Postal Code	Whole number	Postal code	Geographic detail
Region	Text	Regional grouping	Regional performance
Product ID	Text	Product identifier	Product-level analysis
Category	Text	Product category	Category analysis
Sub-Category	Text	Product sub-category	Sub-category profitability
Product Name	Text	Product name	Product ranking/detail
Sales	Decimal number	Sales amount	Revenue KPIs
Quantity	Whole number	Units sold	Volume analysis
Discount	Decimal number	Discount value/rate represented in the source	Discount analysis
Profit	Decimal number	Profit amount	Profitability KPIs

7. Data Preparation & Cleaning

The supplied project documentation states that the dataset was inspected and prepared using Power Query/Power BI transformation capabilities before visualization. The workflow included checking field types, preparing date fields for monthly/yearly analysis, ensuring numeric fields were suitable for aggregation, and creating reusable KPI measures.

Verified preparation activities

- Data structure and field types were checked before visual development.
- Power Query/Power BI transformation capabilities were used to prepare the data.
- Date fields were prepared for monthly and yearly trend analysis.
- Sales, Profit, Quantity and Discount were treated as numeric analytical fields.
- Reusable DAX measures were created for KPI reporting.

Items not independently verified from the raw source

- Exact missing-value counts.
- Exact duplicate-row count.
- Every Power Query step and transformation expression.
- Complete exported PBIX model metadata.

8. Data Model / Power BI Architecture

The project documentation identifies a superstore fact-style table used for transaction-level analysis and a Date table referenced by the documented prior-year profit calculation. The dashboard uses categorical fields such as Region, Category, Segment and Ship Mode together with numeric measures such as Sales and Profit.

Architecture flow

**Superstore source → Power Query preparation → Power BI data model
→ DAX measures → KPI cards & visuals → Interactive dashboard**

Model verification notes

The project evidence supports the use of a 'Superstore' table and a 'Date' table in the documented DAX. However, the full PBIX relationship diagram and every model property were not available as an independently inspectable model artifact in this generation step. Therefore, cardinalities and every relationship are not asserted here.

9. DAX Measures & Calculations

The following formulas are documented in the project README. They are treated as the project's reference DAX, not as a binary export from PBIX.

Total Sales

```
SUM('Superstore'[Sales])
```

Purpose: Sums sales across the current filter context.

Business interpretation: Revenue KPI.

Total Profit

```
SUM('Superstore'[Profit])
```

Purpose: Sums profit across the current filter context.

Business interpretation: Profit KPI.

Total Orders

```
DISTINCTCOUNT('Superstore'[Order ID])
```

Purpose: Counts distinct orders.

Business interpretation: Order volume KPI.

Total Customers

```
DISTINCTCOUNT('Superstore'[Customer ID])
```

Purpose: Counts distinct customers.

Business interpretation: Customer base KPI.

Profit Margin %

$\text{DIVIDE}([\text{Total Profit}], [\text{Total Sales}], 0)$

Purpose: Calculates profit as a share of sales.

Business interpretation: Margin quality KPI.

Average Order Value

$\text{DIVIDE}([\text{Total Sales}], [\text{Total Orders}], 0)$

Purpose: Calculates average sales per distinct order.

Business interpretation: Order-value KPI.

Profit PY

$\text{CALCULATE}([\text{Total Profit}], \text{DATEADD}('Date'[\text{Date}], -1, \text{YEAR}))$

Purpose: Returns prior-year profit using the Date table.

Business interpretation: Supports YoY comparison.

Profit YoY %

$\text{DIVIDE}([\text{Total Profit}] - [\text{Profit PY}], [\text{Profit PY}], 0)$

Purpose: Calculates year-over-year profit change.

Business interpretation: Profit growth KPI.

Customer Sales Rank

```
RANKX(ALL('Superstore'[Customer Name]), [Total Sales], , DESC,  
DENSE)
```

Purpose: Ranks customers by sales.

Business interpretation: Top-customer analysis.

10. KPI / Metrics Analysis

KPI	Dashboard value	What it measures	Why it matters
Total Sales	\$2M	Total revenue represented in the dashboard.	Revenue scale and commercial volume.
Total Profit	\$286K	Total profit represented in the dashboard.	Absolute profit contribution.
Total Orders	5K	Distinct orders represented in the project.	Order volume and activity.
Total Customers	793	Distinct customers represented in the project.	Distinct customer base represented.
Profit Margin	12.5%	Profit as a percentage of sales.	Profitability relative to revenue.
Average Order Value	\$459	Average sales value per order.	Average revenue per order.

Profit YoY	14.2%	Year-over-year profit change shown on the profitability page.	Change in profit compared with the prior year.
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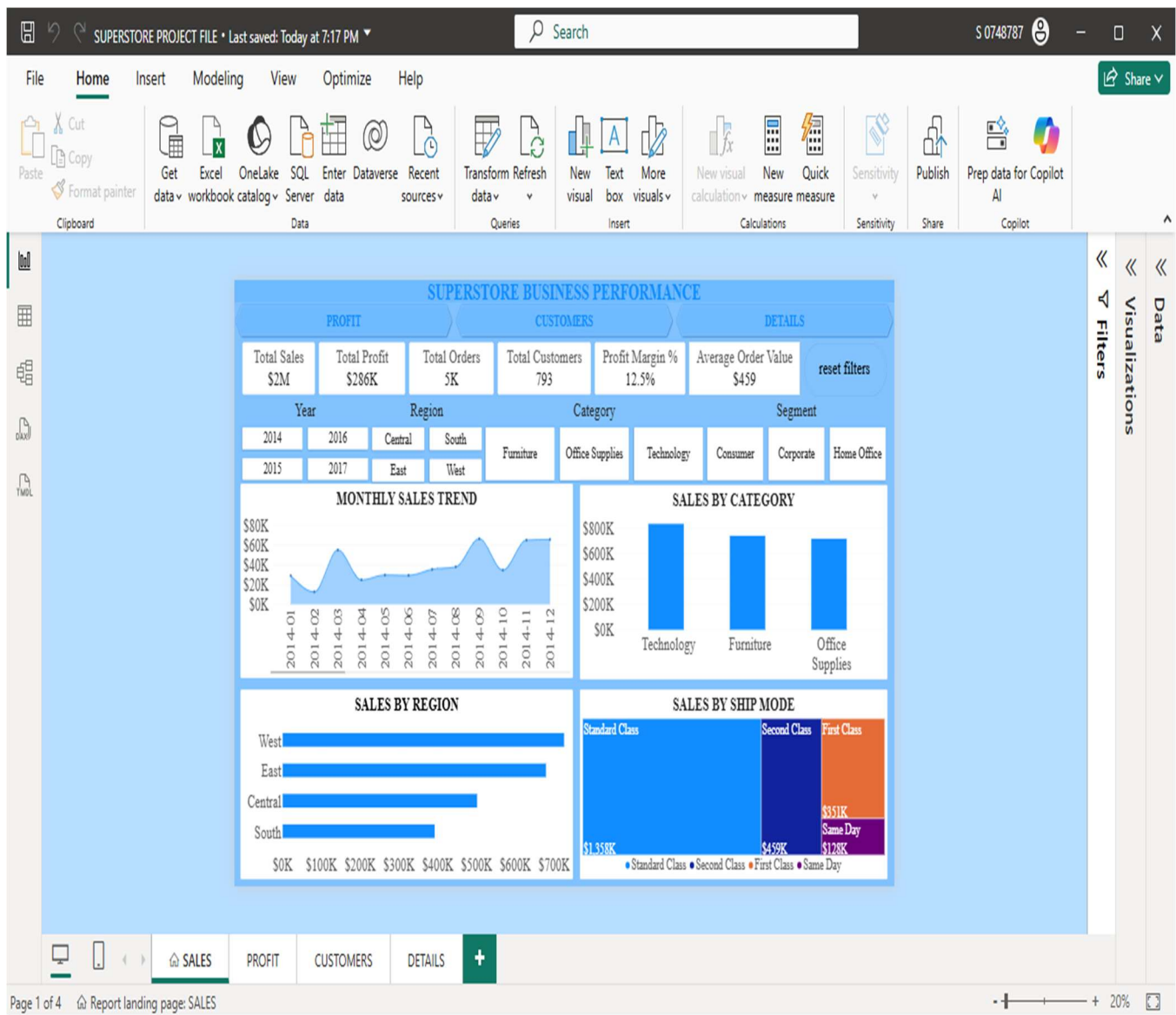
At the dashboard level, approximately \$2M of sales and \$286K of profit correspond to the displayed 12.5% profit margin. The \$459 average order value provides an order-level context for the sales total, while the 14.2% Profit YoY KPI indicates positive year-over-year profit movement in the displayed calculation.

11. Dashboard Design & Navigation

- Four-page structure: Sales, Profitability, Customers and Details.
- Consistent blue theme with white analytical cards.
- KPI cards provide immediate business context.
- Year, Region, Category and Segment slicers support focused analysis.
- Page-navigation tabs connect the four analytical views.
- A Reset Filters button is provided on the Sales page.
- Visual cross-filtering supports interactive exploration.
- Tables use conditional formatting to distinguish positive and negative profit.
- Ranking/top N visuals highlight priority customers and products.

The design is management-oriented: the user can begin with high-level KPIs, move to category/region performance, diagnose profitability, and then inspect customers and products.

12. Dashboard Page 1 — Sales Analysis



Page purpose

The Sales page is the report landing view. It combines high-level KPIs with time, category, regional and shipping analysis so a user can quickly understand overall performance.

Visual-by-visual explanation

Visual	Type	Content	Interpretation
KPI cards	Card visuals	Total Sales, Total Profit, Total Orders, Total Customers, Profit Margin and Average Order Value	Provide an immediate performance snapshot.
Monthly Sales Trend	Time-series/area-style visual	Sales over the displayed monthly period	Shows how revenue changes through time and highlights stronger/weaker periods.
Sales by Category	Column chart	Sales by Technology, Furniture and Office Supplies	Technology is the largest displayed category at about \$836K.
Sales by Region	Horizontal bar chart	Sales in West, East, Central and South	West leads at about \$700K; South is lowest at about \$400K.

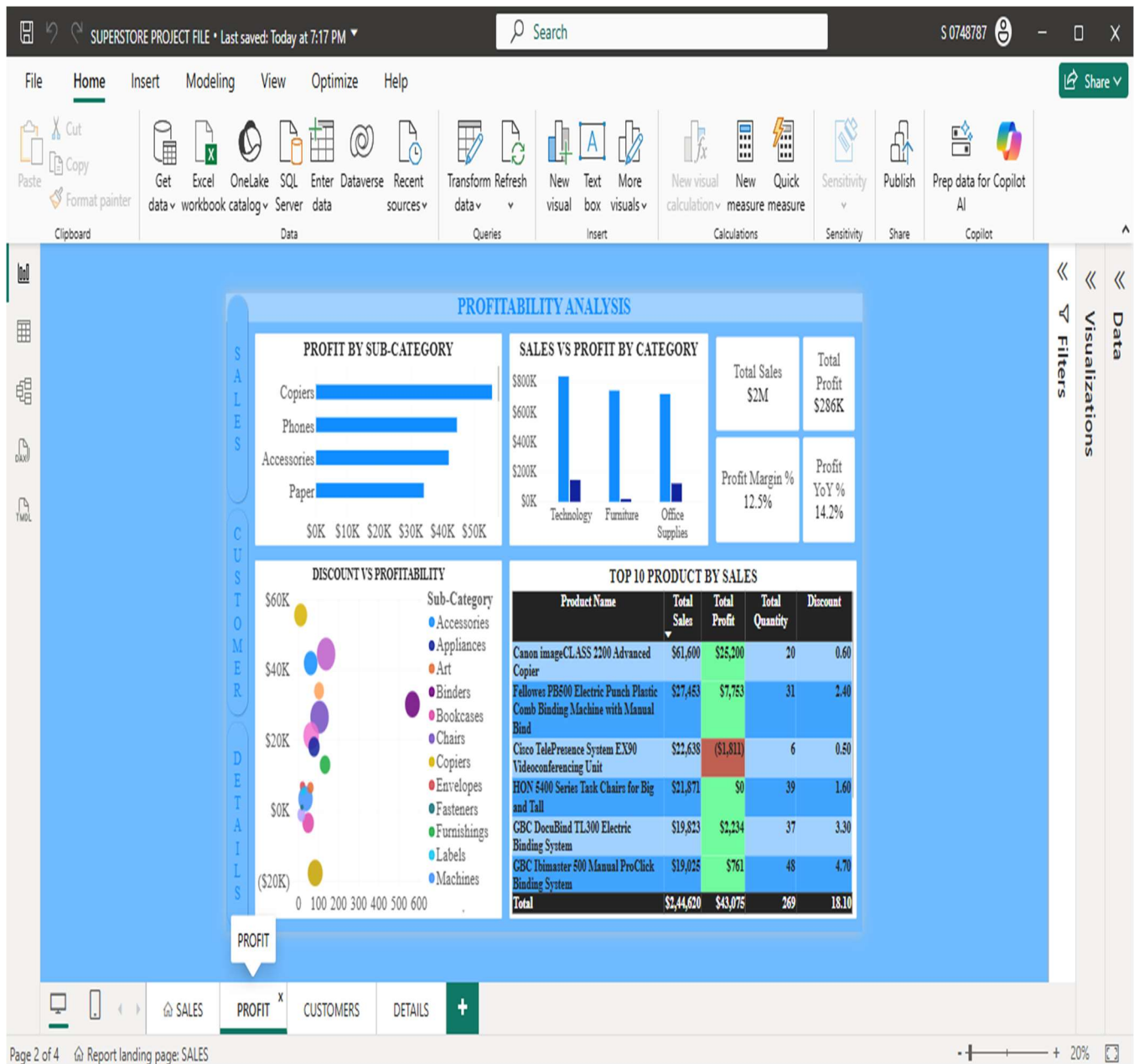
Sales by Ship Mode	Treemap	Sales by shipping mode	Standard Class dominates at about \$1.36M.
Slicers	Slicer controls	Year, Region, Category and Segment	Allow focused analysis without rebuilding visuals.
Reset Filters	Button	Returns the Sales page to the default filter view	Improves usability during interactive analysis.

Sales findings

Dimension	Observed result	Interpretation
Technology	\$836K (~36%)	Largest category displayed by sales.
Furniture	\$742K (~32%)	Second-largest category displayed by sales.
Office Supplies	\$719K (~31%)	Strong sales contributor, close to Furniture.
West	~\$700K	Highest displayed regional sales.

East	~\$650K	The second highest displayed regional sales.
Central	~\$500K	Mid-level contribution among displayed regions.
South	~\$400K	Lowest of the four displayed regions.
Standard Class	~\$1.36M	Dominant shipping mode by sales.

13. Dashboard Page 2 — Profitability Analysis



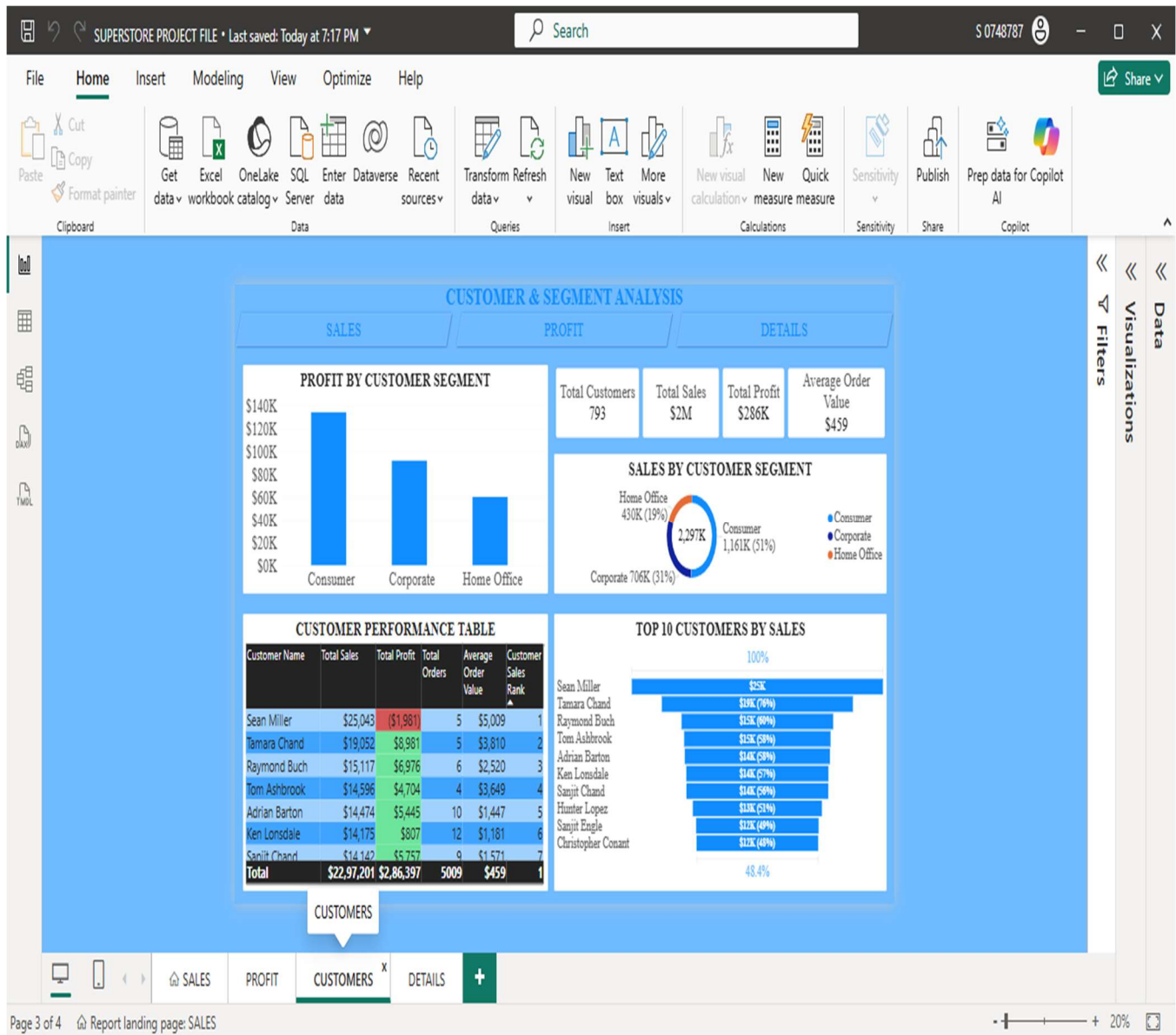
The Profitability page shifts the analysis from revenue volume to financial quality. It combines sub-category profit, category sales versus profit, discount-versus-profitability analysis and product-level results.

Visual	Type	Purpose	Key evidence
Profit by Sub-Category	Horizontal bar chart	Compare profit across sub-categories	Identifies stronger and weaker sub-category contributions.
Sales vs Profit by Category	Column comparison	Contrast sales volume with profit	Technology leads profit at about \$145K; Office Supplies about \$122K; Furniture about \$18K.
Discount vs Profitability	Scatter plot	View discount alongside profitability	Shows why discount should be evaluated together with profit.
Top 10 Product	Table	Review product sales, profit, quantity and discount	Makes product-level positive and negative profit visible.
KPI cards	Cards	Total Sales, Total Profit, Profit Margin, YOY	Provides financial context.

Profitability findings

- Technology is the strongest displayed profit category at approximately \$145K.
- Office Supplies contributes approximately \$122K profit.
- Furniture produces substantial sales but only about \$18K profit in the displayed category comparison, making it the clearest margin-management area.
- The dashboard displays an overall 12.5% profit margin and 14.2% Profit YoY.
- Negative-profit products are visible in the product-level analysis, demonstrating that sales volume alone is not a sufficient performance measure.

14. Dashboard Page 3 — Customer & Segment Analysis



This page evaluates customer value at both segment and individual-customer level.

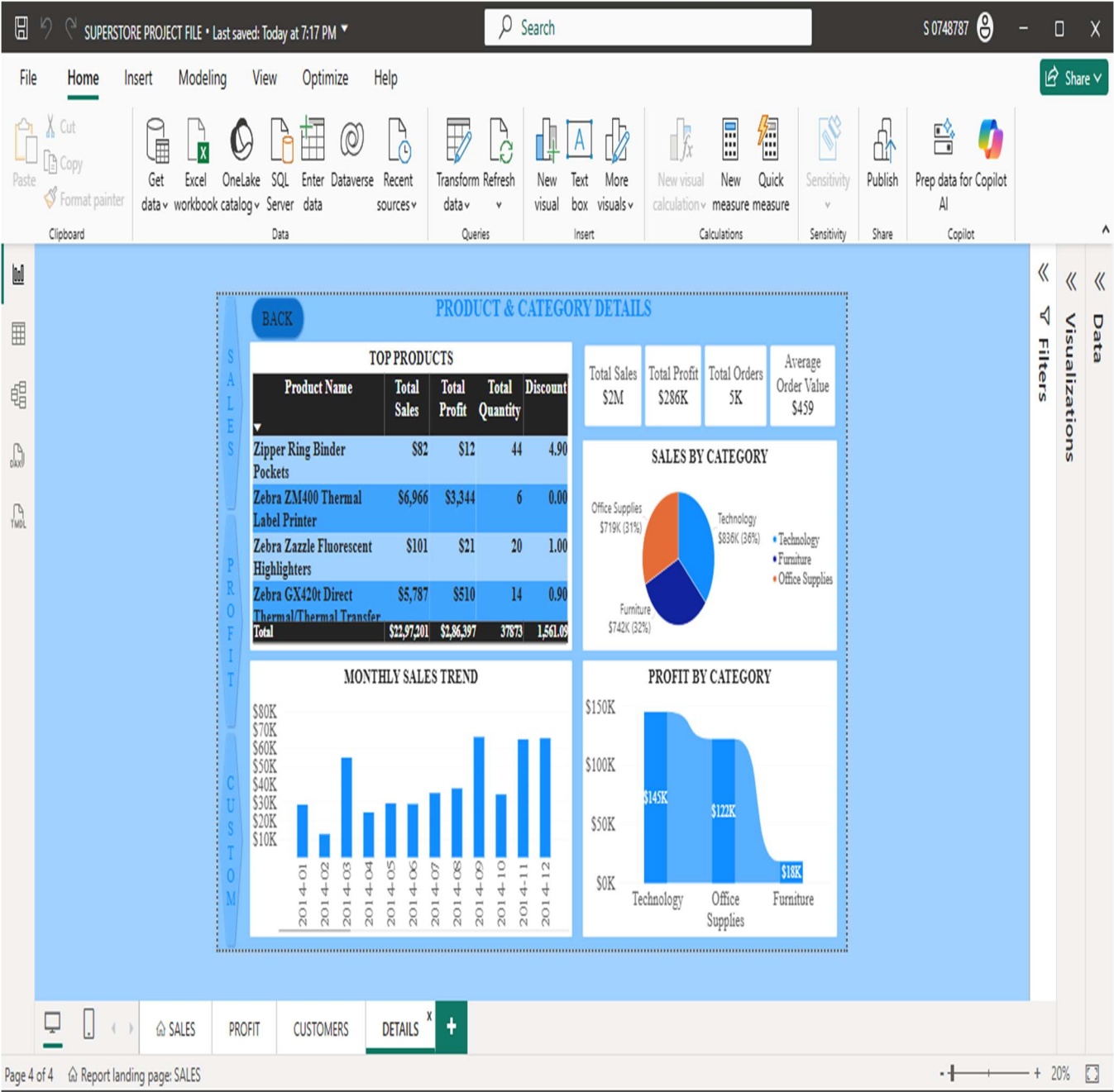
Visual	Type	Purpose	Key evidence
Profit by Customer Segment	Column chart	Compare segment profit	Consumer is the strongest segment by displayed profit.
Sales by Customer Segment	Donut chart	Compare segment sales and share	Consumer \$1.161M (51%), Corporate \$706K (31%), Home Office \$430K (19%).
Customer Performance Table	Table	Review of customer sales, profit, orders, AOV and rank	Shows why customer value should be assessed using profit as well as sales.
Top 10 Customers by Sales	Funnel-style ranking visual	Rank top customers by sales	Used as a customer concentration/ranking view, not a conversion funnel.
KPI cards	Cards	Customers, Sales, Profit and AOV	Provides segment/customer context.

Segment findings

Segment	Sales shown	Interpretation
Consumer	\$1.161M (~51%)	Largest sales segment.
Corporate	\$706K (~31%)	Second-largest segment.
Home Office	\$430K (~19%)	Smaller but meaningful segment.

The visible customer table also demonstrates an important analytical point: a customer can have high sales and still contribute negative profit. One visible example is a customer with approximately \$25K sales and a negative profit contribution.

15. Dashboard Page 4 — Customer & Product Details



The Details page provides a more granular product/category view. It is designed for investigation after a user identifies an important category, product or time pattern on the summary pages.

Visual	Type	Purpose	Key evidence
Top Products	Table	Compare product sales, profit, quantity and discount	Supports product-level decisions.
Sales by Category	Pie chart	Show category sales mix	Technology, Furniture and Office Supplies are compared.
Monthly Sales Trend	Column chart	Review monthly sales	Provides a second detailed time view.
Profit by Category	Category profit chart	Compare profit contribution	Technology is highest; Furniture is much lower in the displayed profit view.
KPI cards	Cards	Sales, Profit, Orders and AOV	Keeps granular analysis anchored to overall performance.
Back button	Navigation button	Return to the previous/main analysis view	Supports report navigation.

Example product-level observations

Product	Sales	Profit	Quantity	Discount
Storex Dura Pro Binders	\$279	\$51	71	7.20
Avery Non-Stick Binders	\$217	\$44	71	6.80
GBC Instant Report Kit	\$165	\$3	50	6.40
Avery Self-Adhesive Photo Pockets for Polaroid Photos	\$143	(\$2)	41	5.90
GBC Standard Recycled Report Covers, Clear	\$259	(\$27)	51	5.90

These examples reinforce the project's central analytical principle: product decisions should consider sales, profit, quantity and discount together. A product can generate revenue while still producing a loss.

16. Key Business Insights

Area	Verified dashboard insight
Overall performance	Approximately \$2M sales and \$286K profit are displayed, with a 12.5% profit margin and 14.2% Profit YoY.
Category	Technology leads displayed sales at about \$836K and displayed profit at about \$145K.
Furniture	Furniture contributes about \$742K sales but only about \$18K profit in the displayed category comparison, indicating a clear profitability-management opportunity.
Office Supplies	Office Supplies contributes about \$719K sales and about \$122K profit, making it a major contributor.
Region	West leads displayed regional sales at about \$700K; South is lowest at about \$400K.
Customer segment	Consumer is the largest segment at about \$1.161M or 51% of displayed sales.

Customers	Top-customer ranking is useful, but the customer table shows why sales alone can hide negative profit contributions.
Products	The top-product table combines sales, profit, quantity and discount, exposing products that sell but lose money.
Discounts	The scatter view makes discount and profitability visible together; the relationship should be investigated without assuming causation.
Shipping	Standard Class dominates displayed sales at about \$1.36M.
Time	The monthly trend shows changes in sales across the 2014–2017 period; the dashboard supports year/month exploration.

17. Business Recommendations

Finding	Recommended action	Expected benefit	Priority
Furniture margin management	Review pricing, procurement cost and discount levels in Furniture and its weaker sub-categories.	Improve profit quality without relying only on sales growth.	High
Protect Technology strength	Maintain availability and competitive pricing for strong Technology products while monitoring margin.	Protect the strongest sales/profit engine.	High
Investigate South region	Compare South product mix, customer mix, discounts and order values with stronger regions.	Identify practical drivers behind the regional gap.	Medium

Retain Consumer customers	Use the largest segment as a priority for retention and cross-selling initiatives.	Protect the largest revenue base.	High
Review loss-making products	Flag negative-profit products for pricing, cost and discount review before increasing promotion.	Reduce avoidable profit leakage.	High
Evaluate discount effectiveness	Assess discount levels alongside profit rather than sales alone.	Improve margin discipline.	Medium
Review shipping economics	Standard Class dominates sales; compare shipping cost and service expectations before changing the mix.	Balance operational cost and customer service.	Medium

18. Limitations & Data Considerations

- The project is descriptive; predictive forecasting is not part of the implemented dashboard.
- The dashboard does not incorporate external competitor, market or macroeconomic data.
- Operational cost drivers beyond the transaction fields are not separately modeled.
- Several dashboard values are rounded (for example, \$2M, \$286K, 5K), so displayed figures should not be treated as exact raw totals.
- The raw source workbook/CSV was not independently mounted during this report-generation step; dataset profile details therefore follow the supplied README/project documentation.
- Full PBIX model metadata and every Power Query transformation step were not independently inspectable here.
- Discount-versus-profitability visuals show association in the displayed data; they do not establish causality.

Interpretation rule used in this report

Where the dashboard provides rounded values, the report preserves the rounded presentation. Where a statement could not be verified from the supplied project evidence, it is explicitly identified rather than inferred as fact.

19. Skills & Tools Demonstrated

Skill / tool	Demonstrated use
Microsoft Power BI Desktop	Dashboard development and interactive reporting
Power Query	Data preparation and transformation
DAX	Reusable measures and analytical calculations
Data Modeling	Use of transactional and date-oriented analytical structures
KPI Development	Sales, profit, orders, customers, margin and AOV
Data Visualization	Charts, tables, KPI cards, scatter and ranking visuals
Business Analysis	Category, region, customer, product, discount and shipping analysis
Dashboard Design	Consistent layout, theme and information hierarchy
Interactive Filtering	Slicers, cross-filtering and reset controls
Business Insight Communication	Translating dashboard patterns into findings and recommendations

20. Conclusion

The Superstore Business Performance & Profitability Analysis project demonstrates an end-to-end Power BI workflow: transactional data preparation, analytical calculation, KPI design, interactive visualization and business interpretation. The four-page dashboard creates a clear progression from overall sales performance to profitability diagnosis, customer/segment analysis and product-level detail.

The analysis shows why sales should not be considered in isolation. Technology is a strong sales and profit driver, Furniture highlights a gap between revenue and profit, Consumer is the largest customer segment, West leads regional sales, and Standard Class dominates shipping sales. At the same time, the product and customer tables reveal negative-profit cases that would be missed by revenue-only analysis.

As a portfolio project, the solution demonstrates practical Data Analyst and Power BI capabilities: preparing data, creating reusable DAX measures, designing an interactive dashboard, analyzing business performance and communicating actionable recommendations.

Portfolio statement: Built an interactive 4-page Power BI dashboard for Superstore business performance analysis, covering sales, profitability, customer segments, regional performance, products, discounts and shipping modes, with KPI cards, slicers, interactive visuals and business recommendations.

Appendix A — KPI Summary Table

Metric	Dashboard value	Interpretation
Total Sales	\$2M	Total revenue represented in the dashboard.
Total Profit	\$286K	Total profit represented in the dashboard.
Total Orders	5K	Distinct orders represented in the project.
Total Customers	793	Distinct customers represented in the project.
Profit Margin	12.5%	Profit as a percentage of sales.
Average Order Value	\$459	Average sales value per order.
Profit YoY	14.2%	Year-over-year profit change shown on the profitability page.

Appendix B — DAX Measures Table

Measure	Reference DAX	Purpose
Total Sales	SUM('Superstore'[Sales])	Sums sales across the current filter context.
Total Profit	SUM('Superstore'[Profit])	Sums profit across the current filter context.
Total Orders	DISTINCTCOUNT('Superstore'[Order ID])	Counts distinct orders.
Total Customers	DISTINCTCOUNT('Superstore'[Customer ID])	Counts distinct customers.
Profit Margin %	DIVIDE([Total Profit], [Total Sales], 0)	Calculates profit as a share of sales.
Average Order Value	DIVIDE([Total Sales], [Total Orders], 0)	Calculates average sales per distinct order.

Profit PY	CALCULATE([Total Profit], DATEADD('Date'[Date], -1, YEAR))	Returns prior-year profit using the Date table.
Profit YoY %	DIVIDE([Total Profit] - [Profit PY], [Profit PY], 0)	Calculates year- over-year profit change.
Customer Sales Rank	RANKX(ALL('Superstore'[Customer Name]), [Total Sales], , DESC, DENSE)	Ranks customers by sales.

Appendix C — Dashboard Visual Inventory

Page	Visual inventory
Sales	6 KPI cards; 4 slicers; Monthly Sales Trend; Sales by Category; Sales by Region; Sales by Ship Mode; Reset Filters; page navigation
Profitability	Profit by Sub-Category; Sales vs Profit by Category; Discount vs Profitability scatter; Top 10 Product table; Total Sales; Total Profit; Profit Margin; Profit YoY
Customers	Profit by Customer Segment; Sales by Customer Segment; Customer Performance Table; Top 10 Customers by Sales ranking/funnel-style visual; KPI cards
Details	Top Products table; Sales by Category; Monthly Sales Trend; Profit by Category; KPI cards; Back button

Appendix D — Dataset / Data Dictionary

Field	Type	Meaning	Use
Row ID	Whole number	Row-level identifier	Record identification
Order ID	Text	Order identifier	Order/order-line analysis
Order Date	Date	Date the order was placed	Time-series analysis
Ship Date	Date	Date the order was shipped	Shipping/time analysis
Ship Mode	Text	Shipping method	Shipping mix analysis
Customer ID	Text	Customer identifier	Customer counting and analysis
Customer Name	Text	Customer name	Customer ranking/detail
Segment	Text	Customer segment	Segment comparison
Country	Text	Country	Geographic context
City	Text	City	Geographic detail

State	Text	State	Geographic detail
Postal Code	Whole number	Postal code	Geographic detail
Region	Text	Regional grouping	Regional performance
Product ID	Text	Product identifier	Product-level analysis
Category	Text	Product category	Category analysis
Sub-Category	Text	Product sub-category	Sub-category profitability
Product Name	Text	Product name	Product ranking/detail
Sales	Decimal number	Sales amount	Revenue KPIs
Quantity	Whole number	Units sold	Volume analysis
Discount	Decimal number	Discount value/rate represented in the source	Discount analysis
Profit	Decimal number	Profit amount	Profitability KPIs

Appendix E — Key Insights Summary

- Overall performance: Approximately \$2M sales and \$286K profit are displayed, with a 12.5% profit margin and 14.2% Profit YoY.
- Category: Technology leads displayed sales at about \$836K and displayed profit at about \$145K.
- Furniture: Furniture contributes about \$742K sales but only about \$18K profit in the displayed category comparison, indicating a clear profitability-management opportunity.
- Office Supplies: Office Supplies contributes about \$719K sales and about \$122K profit, making it a major contributor.
- Region: West leads displayed regional sales at about \$700K; South is lowest at about \$400K.
- Customer segment: Consumer is the largest segment at about \$1.161M or 51% of displayed sales.
- Customers: Top-customer ranking is useful, but the customer table shows why sales alone can hide negative profit contributions.
- Products: The top-product table combines sales, profit, quantity and discount, exposing products that sell but lose money.
- Discounts: The scatter view makes discount and profitability visible together; the relationship should be investigated without assuming causation.
- Shipping: Standard Class dominates displayed sales at about \$1.36M.
- Time: The monthly trend shows changes in sales across the 2014–2017 period; the dashboard supports year/month exploration.

Appendix F — Business Recommendations

Furniture margin management — High priority

Action: Review pricing, procurement cost and discount levels in Furniture and its weaker sub-categories.

Benefit: Improve profit quality without relying only on sales growth.

Protect Technology strength — High priority

Action: Maintain availability and competitive pricing for strong Technology products while monitoring margin.

Benefit: Protect the strongest sales/profit engine.

Investigate South region — Medium priority

Action: Compare South product mix, customer mix, discounts and order values with stronger regions.

Benefit: Identify practical drivers behind the regional gap.

Retain Consumer customers — High priority

Action: Use the largest segment as a priority for retention and cross-selling initiatives.

Benefit: Protect the largest revenue base.

Review loss-making products — High priority

Action: Flag negative-profit products for pricing, cost and discount review before increasing promotion.

Benefit: Reduce avoidable profit leakage.

Evaluate discount effectiveness — Medium priority

Action: Assess discount levels alongside profit rather than sales alone.

Benefit: Improve margin discipline.

Review shipping economics — Medium priority

Action: Standard Class dominates sales; compare shipping cost and service expectations before changing the mix.

Benefit: Balance operational cost and customer service.

Appendix G — GitHub Project Description

Superstore Business Performance & Profitability Analysis — Power BI

An interactive 4-page Power BI dashboard built from the Sample Superstore transactional dataset to analyze sales performance, profitability, customers, products, regions, discounts and shipping modes.

Project at a glance

- Tool: Microsoft Power BI Desktop
- Data preparation: Power Query
- Calculations: DAX
- Dataset: Sample Superstore
- Records: 9,994
- Fields: 21
- Period: 2014–2017
- Dashboard pages: 4
- Focus: Sales, Profitability, Customers & Product Details

Key dashboard KPIs

- Total Sales: \$2M
- Total Profit: \$286K
- Total Orders: 5K
- Total Customers: 793
- Profit Margin: 12.5%
- Average Order Value: \$459

- Profit YoY: 14.2%

Main insights

- Technology leads displayed sales and profit.
- Furniture has strong sales but comparatively low displayed profit.
- West is the strongest displayed sales region.
- Consumer contributes approximately 51% of displayed sales.
- Standard Class is the dominant shipping mode by sales.
- Negative-profit customers/products demonstrate why revenue alone is insufficient.

Skills

Power BI, Power Query, DAX, data preparation, KPI development, data visualization, interactive filtering, dashboard design and business analysis.

Appendix H — Resume Project Description

Superstore Business Performance & Profitability Analysis | Microsoft Power BI

Developed an interactive 4-page Power BI dashboard using Sample Superstore data (9,994 records, 2014–2017), applying Power Query and DAX to build KPI reporting for sales, profit, orders, customers, margin and AOV. Analyzed category, region, customer segment, product, discount and shipping performance and translated findings into business recommendations.

Short version: Built a 4-page Power BI dashboard for Superstore sales and profitability analysis using Power Query, DAX, KPI cards, slicers, ranking visuals and interactive business reporting.

Appendix I — Technical Skills Used

- Microsoft Power BI Desktop
- Power Query
- DAX
- Data preparation and transformation
- Data modeling
- KPI and metric development
- Time-series analysis
- Category and regional analysis
- Customer and segment analysis
- Product and sub-category analysis
- Discount-versus-profitability analysis
- Top-N and ranking analysis
- Interactive slicers and cross-filtering
- Dashboard navigation and usability design
- Business insight generation and recommendation writing